

Tutorial original en ultralytics

[https://colab.research.google.com/github/ultralytics/ultralytics/blob/main/examples/tutorial.ipynb#scrollTo=bpF9-vS\\_DAaf](https://colab.research.google.com/github/ultralytics/ultralytics/blob/main/examples/tutorial.ipynb#scrollTo=bpF9-vS_DAaf)

App para anotar/etiquetar imágenes de nuestra propia base de datos:

<https://app.roboflow.com/login>

```
%pip install ultralytics
import ultralytics
ultralytics.checks()
```

Ultralytics 8.4.150 🚀 Python-3.13.15 torch-2.11.0+cu128 CUDA:0 (Tesla T4, 14913MiB)  
Setup complete ✅ (2 CPUs, 12.7 GB RAM, 47.4/112.6 GB disk)

## ✓ 1. Predict

YOLOv8 may be used directly in the Command Line Interface (CLI) with a `yolo` command for a variety of tasks and modes and accepts additional arguments, i.e. `imgsz=640`. See a full list of available `yolo` [arguments](#) and other details in the [YOLOv8 Predict Docs](#).

```
# Run inference on an image with YOLOv8n
!yolo predict model=yolov8n.pt source='https://ultralytics.com/images/zidane.jpg'
```

Downloading <https://github.com/ultralytics/assets/releases/download/v8.4.0/yolov8n.pt> to 'yolo'...  
Ultralytics 8.4.150 🚀 Python-3.13.15 torch-2.11.0+cu128 CUDA:0 (Tesla T4, 14913MiB)  
YOLOv8n summary (fused): 72 layers, 3,151,904 parameters, 0 gradients, 8.7 GFLOPs

Downloading <https://ultralytics.com/images/zidane.jpg> to 'zidane.jpg': 100% ————— 49.2K  
image 1/1 /content/zidane.jpg: 384x640 2 persons, 1 tie, 6.9ms  
Speed: 318.2ms preprocess, 6.9ms inference, 39.9ms postprocess per image at shape (1, 3, 384, 640)  
Results saved to /content/runs/detect/predict  
💡 Learn more at <https://docs.ultralytics.com/modes/predict>

```
# Run inference on my own image with YOLOv8n (CLI)
!yolo predict model=yolov8n.pt source='https://stilloutriding.com/wp-content/uploads/2025/06/
```

Ultralytics 8.4.150 🚀 Python-3.13.15 torch-2.11.0+cu128 CUDA:0 (Tesla T4, 14913MiB)  
YOLOv8n summary (fused): 72 layers, 3,151,904 parameters, 0 gradients, 8.7 GFLOPs

Downloading <https://stilloutriding.com/wp-content/uploads/2025/06/X1005967-1920x1280.jpg> to 'X1005967-1920x1280.jpg': 100% ————— 1.9M  
image 1/1 /content/X1005967-1920x1280.jpg: 448x640 1 person, 1 bicycle, 8.1ms  
Speed: 35.1ms preprocess, 8.1ms inference, 24.8ms postprocess per image at shape (1, 3, 448, 640)  
Results saved to /content/runs/detect/predict-3  
💡 Learn more at <https://docs.ultralytics.com/modes/predict>

## ✓ 1. Detection

YOLOv8 *detection* models have no suffix and are the default YOLOv8 models, i.e. `yolov8n.pt` and are pretrained on COCO. See [Detection Docs](#) for full details.

```
# Load YOLOv8n, train it on COCO128 for 3 epochs and predict an image with it
from ultralytics import YOLO
```

```
model = YOLO('yolov8n.pt') # load a pretrained YOLOv8n detection model
model.train(data='coco128.yaml', epochs=3) # train the model
model('https://ultralytics.com/images/bus.jpg', save=True) # predict on an image
```

Ultralytics 8.4.150 🚀 Python-3.13.15 torch-2.11.0+cu128 CUDA:0 (Tesla T4, 14913MiB)  
**engine/trainer:** agnostic\_nms=False, amp=True, angle=1.0, augment=False, auto\_augment=randaug

WARNING ⚠ Dataset 'coco128.yaml' images not found, missing path '/content/datasets/coco128/  
Downloading <https://ultralytics.com/assets/coco128.zip> to '/content/datasets/coco128.zip': 100%  
Unzipping /content/datasets/coco128.zip to /content/datasets/coco128...: 100%  
Dataset download success ✅ (0.5s), saved to /content/datasets

Downloading <https://ultralytics.com/assets/Arial.ttf> to '/root/.config/Ultralytics/Arial.ttf'

|    | from         | n  | params | module                             | arguments                            |               |
|----|--------------|----|--------|------------------------------------|--------------------------------------|---------------|
| 0  |              | -1 | 1      | 464                                | ultralytics.nn.modules.conv.Conv     | [3, 16, 3]    |
| 1  |              | -1 | 1      | 4672                               | ultralytics.nn.modules.conv.Conv     | [16, 32, 3]   |
| 2  |              | -1 | 1      | 7360                               | ultralytics.nn.modules.block.C2f     | [32, 32, 3]   |
| 3  |              | -1 | 1      | 18560                              | ultralytics.nn.modules.conv.Conv     | [32, 64, 3]   |
| 4  |              | -1 | 2      | 49664                              | ultralytics.nn.modules.block.C2f     | [64, 64, 3]   |
| 5  |              | -1 | 1      | 73984                              | ultralytics.nn.modules.conv.Conv     | [64, 128, 3]  |
| 6  |              | -1 | 2      | 197632                             | ultralytics.nn.modules.block.C2f     | [128, 128, 3] |
| 7  |              | -1 | 1      | 295424                             | ultralytics.nn.modules.conv.Conv     | [128, 256, 3] |
| 8  |              | -1 | 1      | 460288                             | ultralytics.nn.modules.block.C2f     | [256, 256, 3] |
| 9  |              | -1 | 1      | 164608                             | ultralytics.nn.modules.block.SPPF    | [256, 256, 3] |
| 10 |              | -1 | 1      | 0                                  | torch.nn.modules.upsampling.Upsample | [None, 2, 3]  |
| 11 | [-1, 6]      | 1  | 0      | 0                                  | ultralytics.nn.modules.conv.Concat   | [1]           |
| 12 |              | -1 | 1      | 148224                             | ultralytics.nn.modules.block.C2f     | [384, 128, 3] |
| 13 |              | -1 | 1      | 0                                  | torch.nn.modules.upsampling.Upsample | [None, 2, 3]  |
| 14 | [-1, 4]      | 1  | 0      | 0                                  | ultralytics.nn.modules.conv.Concat   | [1]           |
| 15 |              | -1 | 1      | 37248                              | ultralytics.nn.modules.block.C2f     | [192, 64, 3]  |
| 16 |              | -1 | 1      | 36992                              | ultralytics.nn.modules.conv.Conv     | [64, 64, 3]   |
| 17 | [-1, 12]     | 1  | 0      | 0                                  | ultralytics.nn.modules.conv.Concat   | [1]           |
| 18 |              | -1 | 1      | 123648                             | ultralytics.nn.modules.block.C2f     | [192, 128, 3] |
| 19 |              | -1 | 1      | 147712                             | ultralytics.nn.modules.conv.Conv     | [128, 128, 3] |
| 20 | [-1, 9]      | 1  | 0      | 0                                  | ultralytics.nn.modules.conv.Concat   | [1]           |
| 21 |              | -1 | 1      | 493056                             | ultralytics.nn.modules.block.C2f     | [384, 256, 3] |
| 22 | [15, 18, 21] | 1  | 897664 | ultralytics.nn.modules.head.Detect | [80, 16, 1]                          |               |

Model summary: 129 layers, 3,157,200 parameters, 3,157,184 gradients, 8.9 GFLOPs

Transferred 355/355 items from pretrained weights  
Freezing layer 'model.22.dfl.conv.weight'  
**AMP:** running Automatic Mixed Precision (AMP) checks...  
**AMP:** downloading yolov8n.pt for AMP checks (one-time, not used for training)...  
Downloading <https://github.com/ultralytics/assets/releases/download/v8.4.0/yolov8n.pt> to 'weights/yolov8n.pt'  
**AMP:** checks passed ✅  
**train:** Fast image access ✅ (ping: 0.0±0.0 ms, read: 1038.4±508.3 MB/s, size: 46.0 KB)  
**train:** Scanning /content/datasets/coco128/labels/train2017... 126 images, 2 backgrounds, 0 coco  
**train:** New cache created: /content/datasets/coco128/labels/train2017.cache  
**augmentations:** Blur(p=0.01, blur\_limit=(3, 7)), MedianBlur(p=0.01, blur\_limit=(3, 7)), ToGrayscale(p=0.01)  
**val:** Fast image access ✅ (ping: 0.0±0.0 ms, read: 479.9±93.5 MB/s, size: 56.4 KB)  
**val:** Scanning /content/datasets/coco128/labels/train2017.cache... 126 images, 2 backgrounds, 0 coco  
**optimizer:** 'optimizer=auto' found, ignoring 'lr0=0.01' and 'momentum=0.937' and determining best values  
**optimizer:** AdamW(lr=0.000119, momentum=0.9) with parameter groups 57 weight(decay=0.0), 64 w  
Plotting labels to /content/runs/detect/train/labels.jpg...  
Using 128 train, 128 val images for fraction=1.0 at imgs=640  
Using 2 dataloader workers  
Logging results to /content/runs/detect/train  
Starting training for 3 epochs...

| Epoch | GPU_mem | box_loss | cls_loss | dfl_loss | Instances | Size     |
|-------|---------|----------|----------|----------|-----------|----------|
| 1/3   | 2.25G   | 1.174    | 1.530    | 1.335    | 173       | 640x1000 |

```
# Load YOLOv8n, train it on COCO128 for 3 epochs and predict an image with it -- Using my own
from ultralytics import YOLO

model = YOLO('yolov8n.pt') # load a pretrained YOLOv8n detection model
model.train(data='coco128.yaml', epochs=3) # train the model
```

```
model('https://stilloutriding.com/wp-content/uploads/2025/06/X1005967-1920x1280.jpg', save=True)
```

```
'snowboard', 32: 'sports ball', 33: 'kite', 34: 'baseball bat', 35: 'baseball glove', 36:
'skateboard', 37: 'surfboard', 38: 'tennis racket', 39: 'bottle', 40: 'wine glass', 41:
'cup', 42: 'fork', 43: 'knife', 44: 'spoon', 45: 'bowl', 46: 'banana', 47: 'apple', 48:
'sandwich', 49: 'orange', 50: 'broccoli', 51: 'carrot', 52: 'hot dog', 53: 'pizza', 54:
'donut', 55: 'cake', 56: 'chair', 57: 'couch', 58: 'potted plant', 59: 'bed', 60: 'dining
table', 61: 'toilet', 62: 'tv', 63: 'laptop', 64: 'mouse', 65: 'remote', 66: 'keyboard', 67:
'cell phone', 68: 'microwave', 69: 'oven', 70: 'toaster', 71: 'sink', 72: 'refrigerator',
73: 'book', 74: 'clock', 75: 'vase', 76: 'scissors', 77: 'teddy bear', 78: 'hair drier', 79:
'toothbrush'}
```

```
obb: None
```

```
orig_img: array([[ 91, 104,  88],
 [ 90, 101,  85],
 [ 99, 110,  94],
```

```
...,
 [243, 229, 210],
 [245, 231, 209],
 [245, 228, 207]],
```

```
[[ 84,  95,  79],
 [ 90, 101,  85],
 [ 94, 105,  89],
```

```
...,
 [240, 226, 208],
 [242, 227, 208],
 [244, 227, 206]],
```

```
[[ 79,  89,  76],
 [ 82,  92,  79],
 [ 80,  90,  77],
```

```
...,
 [245, 229, 213],
 [249, 231, 214],
 [251, 231, 213]],
```

```
...,
```

```
[[ 37,  36,  40],
 [ 33,  34,  38],
 [ 32,  32,  38],
```

```
...,
 [ 40,  43,  47],
 [ 40,  41,  45],
 [ 38,  39,  43]],
```

```
[[ 21,  17,  22],
 [ 26,  25,  29],
 [ 31,  32,  36],
```

```
...,
 [ 38,  41,  45],
 [ 40,  41,  45],
 [ 41,  42,  46]],
```

```
[[ 24,  20,  25],
 [ 19,  15,  20],
 [ 13,  12,  16],
```

```
...,
 [ 35,  38,  42],
 [ 30,  30,  30]]
```

